

shady aspects of subjective valuation procedures—the reluctance of managers to mark to market in down markets.

In many cases, we expect the true illiquid asset returns to be autocorrelated as well.¹⁴ Illiquid asset markets—like real estate, private equity, timber plantations, and infrastructure—are markets where information is not available to all participants, information does not spread rapidly, and capital cannot be immediately deployed into new investments. Informationally inefficient markets with slow-moving capital are characterized by persistent returns.¹⁵

3. Unsmoothing is an art.

The unsmoothing example in equations (13.1) and (13.2) uses the simplest possible autocorrelated process, an AR(1), to describe reported returns. Many illiquid assets have more than first-order lag effects. Real estate, for example, has a well-known fourth-order lag working with quarterly data arising from many properties being reappraised only annually.¹⁶ A good unsmoothing procedure takes a time-series model that fits the reported return data well and then with a general transfer function assumption, the filter for true returns in equation (13.2) becomes a very complicated function of present and past lagged observed returns.¹⁷ Doing this properly requires good statistical skills. It also requires underlying economic knowledge of the structure of the illiquid market to interpret what is a reasonable lag structure and to judge how much unsmoothing is required.

close to the water or far, or whether the house has two stories or one. These are called *hedonic* adjustments. These methods have been applied to create indexes in other illiquid markets, like art (Goetzmann (1993) and Moses and Mei (2002)), stamps (Dimson and Späenkers (2011)), and wine (Krasker (1979) and Masset and Weisskopf (2010)). The aggregation process in constructing indexes of illiquid asset returns induces further smoothing. Indexes combine many individual indications of value, either market transactions or appraised values, and typically the values are appraised at different points throughout the year. Note that if $\phi = 0$, then equations (2) and (3) coincide and unsmoothed returns are exactly the same as reported returns. Figure 13.2, which shows the effects of infrequent observations, is produced with a year-on-year autocorrelation of 0.4.

¹⁴When the true returns are autocorrelated, the horizon matters in stating volatilities, correlations, and Sharpe ratios. From point 1, the means are unaffected. See Lo (2002) for formulas to convert the risk measures for different horizons.

¹⁵See Duffie (2010) and Duffie and Strulovici (2012).

¹⁶This is noted in the seminal Geltner (1991) and Ross and Zisler (1991) papers.

¹⁷We want an ARMA(p, q) model, which captures the effect of p lagged autocorrelated terms (the “AR” effect for p lags) and where innovations to those returns in past periods continue to have an effect on present returns. The latter are referred to as moving average terms (the “MA” effects for q lags). Both Geltner (1991) and Ross and Zisler (1991) consider richer time-series processes than just an AR(1). Okunev and White (2003) and Getmansky, Lo, and Makarov (2004) develop unsmoothing algorithms to hedge fund returns with higher-order autocorrelation corrections.